THE POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY 1

FINAL REPORT

Business Intelligence

Topic: Retail Revenue Analytics BI System

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Hà Nội - 2026

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Introduction

1. Reasons for Choosing the Topic

1.1. The Data Explosion Context in the Retail Industry

In the era of the Fourth Industrial Revolution (Industry 4.0), the retail industry has transformed from traditional store models to omnichannel business models. This transformation has resulted in the generation of transaction data at extremely high speed (Velocity) and in massive amounts (Volume). In retail chains such as Superstore, hundreds of transactions are recorded every second, including detailed information about customers, product categories, geographic locations, and associated promotional programs.

1.2. Challenges of Traditional Data Structures (Flat Tables)

Previously, data was commonly stored in the form of a "Flat Table," where all information was concentrated in a single table. However, as the scale of data grew to millions of records, this structure revealed several critical weaknesses. Customer and product information had to be repeated thousands of times, leading to significant data redundancy and wasting storage resources. In addition, querying and calculating metrics such as total revenue or profit from an enormous table required substantial CPU and RAM resources, causing reports to take a long time to generate and display. Furthermore, maintaining the data became increasingly difficult because any changes to product names or customer addresses had to be updated across millions of rows, which easily resulted in inconsistencies and errors.

1.3. The Need for an Advanced Business Intelligence (BI) System

The topic "Business Intelligence System for Retail Revenue Analysis" was chosen to address one of the most important challenges in modern business management: how to transform large volumes of raw data into meaningful business insights that support decision-making. In today's competitive retail environment, companies continuously generate massive amounts of transactional data from sales activities, customer interactions, inventory management, and promotional campaigns. However, without a specialized Business Intelligence (BI) system, most of this data remains fragmented and underutilized, making it difficult for managers to gain a comprehensive understanding of business performance.

The development of a modern Data Warehouse therefore plays a crucial role not only from a technical perspective but also from a strategic business perspective. By integrating and organizing data from multiple sources into a centralized system, businesses can perform in-depth analysis more efficiently and accurately. A BI system allows managers to identify seasonal purchasing trends, understand customer behavior patterns, and predict future demand based on historical data. In addition, it provides valuable insights into the effectiveness of discount programs and marketing campaigns, helping businesses allocate resources more effectively and maximize profits. Furthermore, analyzing real consumption data enables companies to optimize inventory management and improve supply chain operations, thereby reducing operational costs and enhancing overall business efficiency. Through these capabilities, a Business Intelligence system becomes an essential tool that supports data-driven decision-making and sustainable growth in the retail industry.

2. Research Objectives

The study focuses on three key objectives, representing the three main stages of a professional data analytics project.

2.1. Designing a Data Warehouse Based on the Star Schema Model

The first objective of this study is to restructure data from an operational processing model (OLTP) into an analytical processing model (OLAP) in order to support efficient business analysis and decision-making. To achieve this goal, the research focuses on the implementation of the Star Schema model, one of the most widely used data warehouse architectures in Business Intelligence systems. In this model, the Fact Table is positioned at the center and stores quantitative metrics such as sales revenue, profit, quantity sold, and transaction values, while surrounding Dimension Tables contain descriptive information related to customers, products, locations, and time. This structure helps simplify analytical queries and significantly improves reporting performance.

In addition, the study aims to optimize database relationships through the use of Surrogate Keys instead of natural keys. By applying surrogate keys, the system can increase the speed of table joins, reduce dependency on operational data structures, and maintain the integrity of historical data even when changes occur in the original source systems. As a result, the designed Data Warehouse can provide a stable, scalable, and high-performance foundation for advanced business analytics.

2.2. Building and Implementing the ETL (Extract – Transform – Load) Process Using SQL

The second objective is to implement the process of data migration and data cleaning, which is considered one of the most critical and time-consuming stages in a data engineering project, often accounting for up to 70% of the total workload. This study focuses on building a complete ETL (Extract – Transform – Load) workflow using SQL to ensure that data can be processed efficiently and accurately before being stored in the Data Warehouse.

In the Extract phase, data is collected from multiple sources such as CSV files, Excel spreadsheets, and operational databases, then transferred into a Staging Area for temporary storage and preliminary processing. During the Transform phase, advanced SQL functions and algorithms are applied to clean and standardize the data. This includes handling missing or incorrect values, removing duplicate records, standardizing date and time formats, and generating derived attributes required for business analysis. These transformation processes help improve data consistency, reliability, and analytical quality. Finally, in the Load phase, the cleaned and transformed data is inserted into the final Data Warehouse structure, including Fact Tables and Dimension Tables, making the data ready for reporting, visualization, and strategic business analysis.

2.3. Building a Multi-Dimensional Reporting Dashboard on the Power BI Platform

The final objective of the study is to present data in a visual, interactive, and user-friendly manner through the development of a multi-dimensional dashboard using Power BI. In modern Business Intelligence systems, dashboards play a crucial role in transforming complex datasets into meaningful visual insights that support faster and more accurate decision-making. Therefore, this study focuses not only on data analysis but also on creating an effective user experience for business managers and decision-makers.

One of the primary goals is to develop core business KPIs by using DAX (Data Analysis Expressions) to calculate important indicators such as Net Revenue, Profit Margin, Sales Performance, and Growth Rate. These metrics provide managers with a comprehensive overview of business operations and help evaluate organizational performance over time. In addition, the dashboard is designed with interactive visualization features, including slicers, filters, and drill-down functions, allowing users to explore data at multiple levels of detail. Managers can analyze information from a global business overview down to specific provinces or regions, and from yearly trends to individual daily transactions.

Furthermore, the dashboard is intended to support strategic decision-making by highlighting revenue “hotspots,” identifying underperforming business areas, and detecting changes in customer purchasing behavior. Through intuitive charts, graphs, and dynamic reports, the system enables businesses to quickly recognize opportunities and challenges, thereby allowing timely interventions and more effective business strategies. Ultimately, the Power BI dashboard serves as a powerful analytical tool that enhances data-driven management and improves operational efficiency in the retail industry.

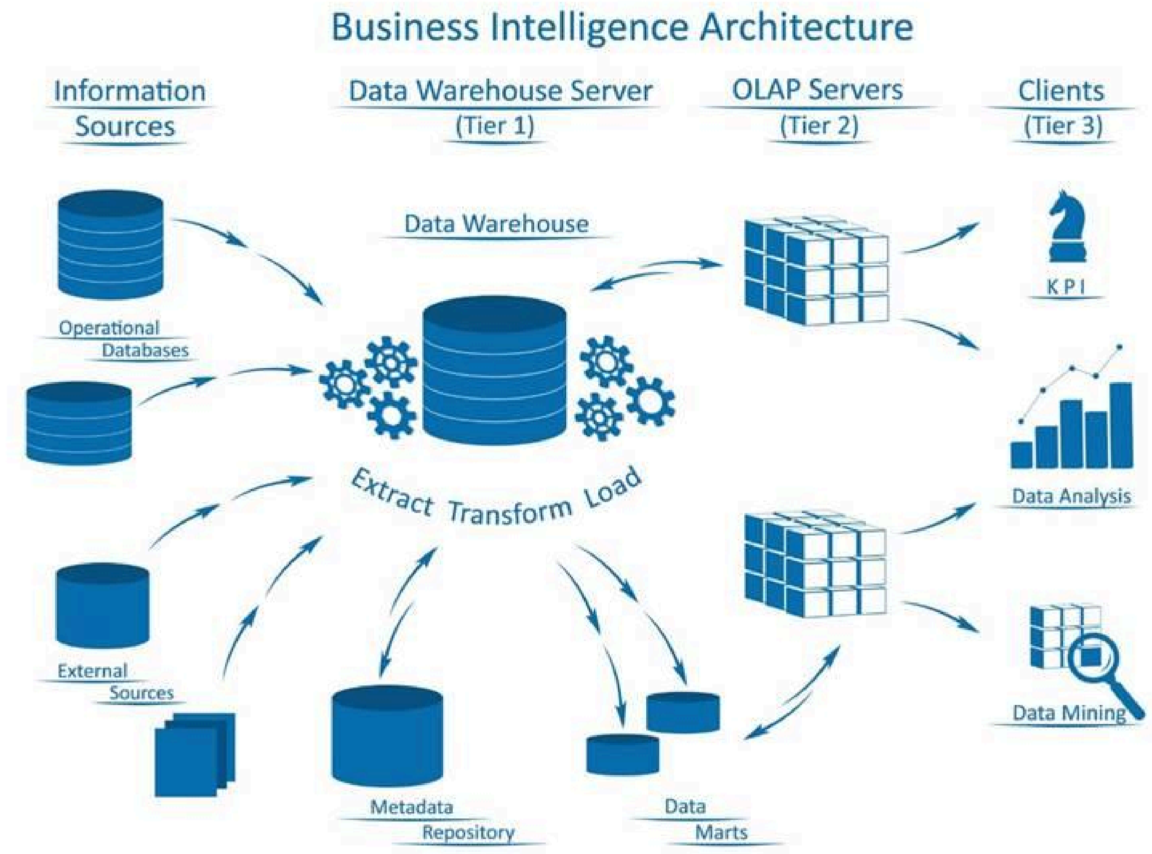
I. OVERVIEW OF BUSINESS INTELLIGENCE SYSTEMS AND DATA WAREHOUSING

1. The Concept of Business Intelligence (BI) and Its Importance

Business Intelligence (BI) is not simply a reporting software solution, but a comprehensive technology ecosystem that includes processes, architectures, and data management tools designed to support business analysis and decision-making. The core objective of BI is to transform raw data into meaningful information and ultimately into actionable insights that can create business value. In the context of modern enterprises, especially in the retail industry, BI has become an essential component for improving competitiveness and operational efficiency in an increasingly data-driven environment.

In this project, the BI system functions as the analytical "brain" of the organization, enabling businesses to process and interpret large volumes of transactional data effectively. One of its most important roles is breaking down data silos by integrating information from multiple sources into a unified and centralized view. This integration allows managers to access consistent and accurate information without relying on fragmented datasets from separate departments or systems.

Furthermore, Business Intelligence supports evidence-based decision-making by replacing intuition-based management with precise metrics and analytical reports related to revenue, profit, customer behavior, and sales performance. Through dashboards, KPIs, and advanced analytical tools, managers can identify trends, evaluate business performance, and make strategic decisions with greater confidence and accuracy. In addition, BI systems contribute significantly to operational optimization by detecting inefficiencies within the supply chain, identifying products with excessive inventory levels, and analyzing areas where business performance is underachieving. As a result, organizations can improve resource allocation, reduce operational costs, and respond more quickly to market changes. Ultimately, Business Intelligence serves as a strategic foundation that helps enterprises enhance competitiveness, improve decision-making quality, and achieve sustainable growth.



2. Data Warehouse Architecture

A Data Warehouse (DWH) is the backbone of a Business Intelligence system, serving as the centralized repository for storing and analyzing large volumes of business data. In this project, the Data Warehouse architecture is designed with a clear separation between the Online Transaction Processing (OLTP) system and the Online Analytical Processing (OLAP) system. This separation is essential because operational systems are optimized for fast transaction processing, while analytical systems are designed to support complex queries, reporting, and large-scale data analysis.

By isolating OLTP and OLAP environments, the system can ensure that daily business operations such as sales transactions, customer updates, and inventory management are not affected by heavy analytical workloads. At the same time, the OLAP system provides a stable and high-performance environment for generating reports, analyzing trends, and supporting strategic decision-making. This architecture also improves scalability, data consistency, and query performance, making it more suitable for handling the growing volume of retail data in modern Business Intelligence applications.

2.1. Characteristics of a Data Warehouse

A Data Warehouse possesses several fundamental characteristics that distinguish it from traditional operational databases. First, it is subject-oriented, meaning that data is organized around major business subjects such as Sales, Customers, and Products rather than around specific software functions or application processes. This approach allows businesses to analyze information more effectively from a managerial and strategic perspective.

Second, a Data Warehouse is integrated, which means that data collected from multiple sources is standardized before being loaded into the system. Different measurement units, date formats, naming conventions, and coding systems are unified to ensure consistency and accuracy across the entire warehouse. This integration process is essential for generating reliable analytical reports and supporting data-driven decision-making.

Another important characteristic is its time-variant nature. Unlike operational systems that mainly store current transactional data, a Data Warehouse maintains historical data over long periods, often spanning multiple years. This enables organizations to analyze trends, evaluate business growth over time, and compare performance across different periods. Historical data storage is especially valuable for forecasting, strategic planning, and long-term business analysis.

Finally, a Data Warehouse is considered non-volatile, meaning that once data has been loaded into the warehouse, it is generally not modified or deleted through regular business transactions. This ensures data stability and consistency, which are critical for generating accurate management reports and maintaining trust in analytical results.

2.2. The Role of System Separation

Separating the Data Warehouse from the operational system provides several significant advantages for both system performance and business operations. One of the most important benefits is the protection of operational performance. Complex analytical queries that involve aggregating and processing millions of records are executed within the Data Warehouse environment rather than on the transactional system. As a result, daily business operations such as sales transactions at retail counters can continue smoothly without being slowed down or interrupted by heavy analytical workloads.

In addition, the separation allows each system to be optimized for its specific purpose. Operational systems primarily focus on fast and efficient write operations because they continuously process real-time transactions such as orders, payments, and inventory updates. In contrast, the Data Warehouse is optimized for read performance and analytical processing, enabling faster report generation, multidimensional analysis, and large-scale data retrieval. This architectural separation improves scalability, enhances system reliability, and creates a more efficient environment for both operational processing and strategic business analysis

3. Star Schema Model

In this project, the Star Schema model was selected as the primary storage structure because of its simplicity, scalability, and exceptional efficiency in performing multidimensional analytical queries. The Star Schema is one of the most widely adopted data modeling techniques in Business Intelligence and Data Warehouse systems due to its ability to optimize query performance while maintaining a structure that is easy for analysts and business users to understand. Its architecture consists of a central Fact Table connected directly to multiple surrounding Dimension Tables, forming a structure that visually resembles a star.

3.1. Fact Table

At the center of the Star Schema is the **Fact_Sales** table, which stores quantitative data generated from business transactions. This table contains measurable numerical values such as Sales, Profit, Quantity, and Discount, all of which are essential for analytical reporting and business performance evaluation. Each record in the Fact Table represents a specific business event or transaction, making it the core component for calculating KPIs and generating analytical insights.

The structure of the Fact Table includes foreign keys that establish relationships with all surrounding Dimension Tables. These relationships allow the system to connect transactional data with descriptive business information such as customer details, product categories, geographic regions, and transaction dates. Because the Fact Table typically contains a very large volume of records, its design focuses on optimizing storage efficiency and query performance for analytical operations.

3.2. Dimension Tables

Surrounding the Fact Table are the Dimension Tables, which provide contextual information for the numerical data stored in the Fact Table. These tables answer important business questions such as “Who?”, “What?”, “Where?”, and “When?”. For example, the **Dim_Product** table contains detailed information about products, including product names, categories, and brands, while the **Dim_Customer** table stores customer-related attributes such as customer segments, demographics, and geographic information.

Unlike the Fact Table, Dimension Tables mainly contain descriptive and textual data. Their records are generally smaller in volume and change less frequently over time. Dimension Tables play a crucial role in enabling multidimensional analysis because they allow users to filter, group, and drill down into data from different business perspectives. This contextual structure makes reports more meaningful and easier to interpret for decision-makers.

3.3. Advantages of the Star Schema in the Project

The implementation of the Star Schema provides several important advantages for this project. First, it significantly improves query performance by minimizing the number of complex table joins required during analytical processing. As a result, Power BI can load dashboards, charts, and reports much faster, even when working with large datasets containing millions of records.

Second, the Star Schema is highly intuitive and user-friendly. Its simple structure allows report developers and business analysts to easily select attributes from Dimension Tables and perform operations such as slicing, dicing, filtering, and drilling down into the data. This enhances the efficiency of report creation and makes analytical exploration more accessible to non-technical users.

Finally, the Star Schema is fully compatible with DAX (Data Analysis Expressions), the calculation language used in Power BI. DAX performs most effectively when working with a well-structured Star Schema because relationships between Fact and Dimension Tables are clearly defined and optimized for analytical calculations. This compatibility enables the creation of advanced KPIs, dynamic measures, and interactive dashboards that support accurate and efficient business analysis.

II. SOURCE DATA ANALYSIS AND DATA WAREHOUSE DESIGN

1. Retail Dataset Analysis

The Superstore dataset provides a comprehensive overview of omnichannel retail operations, including customer transactions, product information, sales performance, and geographic distribution. Before designing the Data Warehouse, it is essential to classify the source data into logical information groups in order to define an efficient and scalable warehouse structure. This analysis stage helps identify the relationships between business entities and determines how the data should be organized within Fact Tables and Dimension Tables for analytical purposes.

1.1. Customer Information Group (Customer Profile)

This group focuses on the customers who perform purchasing activities within the retail system. The attributes in this group include:

- Customer_ID: A unique identifier assigned to each customer, used as the primary key for establishing relationships between tables.
- Customer_Name: The full name of the customer, used for identification purposes in detailed reports.
- Segment: Customer segment classification (Consumer, Corporate, Home Office). This is an important attribute for analyzing purchasing behavior across different target customer groups.

Customer_ID	Customer_Name	Segment
EH-13945	Eric Hoffmann	Consumer
TB-21520	Tracy Blumstein	Consumer
TB-21520	Tracy Blumstein	Consumer
TB-21520	Tracy Blumstein	Consumer
TB-21520	Tracy Blumstein	Consumer
TB-21520	Tracy Blumstein	Consumer
TB-21520	Tracy Blumstein	Consumer
MA-17560	Matt Abelman	Home Off...
GH-14485	Gene Hale	Corporate
GH-14485	Gene Hale	Corporate
SN-20710	Steve Nguyen	Home Off...
SN-20710	Steve Nguyen	Home Off...
SN-20710	Steve Nguyen	Home Off...
SN-20710	Steve Nguyen	Home Off...
LC-16930	Linda Cazamias	Corporate
RA-19885	Ruben Ausman	Corporate
ES-14080	Erin Smith	Corporate
ON-18715	Odella Nelson	Corporate
ON-18715	Odella Nelson	Corporate

1.2. Product Information Group (Product Hierarchy)

Product data is organized in a hierarchical structure, allowing managers to generate reports from a general overview down to detailed product-level analysis:

- Category: Major product categories (Technology, Furniture, Office Supplies).
- Sub_Category: Subcategories of products (Phones, Chairs, Paper, etc.).
- Product_Name & Product_ID: Detailed information about each specific product item. This hierarchical structure supports Drill-down operations in Power BI, allowing users to navigate from summary-level data to more detailed analysis.

Category	Sub_Category	Product_Name
Technology	Accessories	Imation 8GB Mini TravelDrive USB 2.0 Flash Drive
Furniture	Bookcases	Riverside Palais Royal Lawyers Bookcase, Royal...
Office Supplies	Binders	Avery Recycled Flexi-View Covers for Binding S...
Furniture	Furnishings	Howard Miller 13-3/4" Diameter Brushed Chrom...
Office Supplies	Envelopes	Poly String Tie Envelopes
Office Supplies	Art	BOSTON Model 1800 Electric Pencil Sharpeners,...
Office Supplies	Art	Lumber Crayons
Office Supplies	Paper	Easy-staple paper
Technology	Phones	GE 30524EE4
Fu Technology	Furnishings	Electrix Architect's Clamp-On Swing Arm Lamp, ...
Office Supplies	Envelopes	#10-4 1/8" x 9 1/2"" Premium Diagonal Seam En...
Furniture	Bookcases	Atlantic Metals Mobile 3-Shelf Bookcases, Custo...
Furniture	Chairs	Global Fabric Manager's Chair, Dark Gray
Technology	Phones	Plantronics HL10 Handset Lifter
Technology	Phones	Panasonic Kx-TS550
Office Supplies	Storage	Eldon Base for stackable storage shelf, platinum
Office Supplies	Storage	Advantus 10-Drawer Portable Organizer, Chro...
Technology	Accessories	Verbatim 25 GB 6x Blu-ray Single Layer Recorda...
Office Supplies	Binders	Wilson Jones Leather-Like Binders with DublLock...

1.3. Geographic Information Group (Geographic Context)

This information group provides the spatial and geographic context in which business transactions occur. It includes multiple hierarchical levels such as Country, Region, State, City, and Postal_Code. These attributes enable businesses to analyze sales performance and customer behavior across different geographic areas and administrative levels.

The geographic information group also serves as the foundation for building map visualizations and location-based analytical reports in Power BI. By integrating geographic data into the Data Warehouse, managers can identify regional sales trends, compare business performance between locations, and detect high-performing or underperforming markets. Furthermore, map-based visualizations help decision-makers gain intuitive insights into geographic distribution patterns, supporting more effective market expansion strategies and regional business planning.

Country	City	State	Postal_Code	Region
United States	Los Angeles	California	90049	West
United States	Philadelphia	Pennsyl...	19140	East
United States	Philadelphia	Pennsyl...	19140	East
United States	Philadelphia	Pennsyl...	19140	East
United States	Philadelphia	Pennsyl...	19140	East
United States	Philadelphia	Pennsyl...	19140	East
United States	Philadelphia	Pennsyl...	19140	East
United States	Houston	Texas	77095	Central
United States	Richardson	Texas	75080	Central
United States	Richardson	Texas	75080	Central
United States	Houston	Texas	77041	Central
United States	Houston	Texas	77041	Central
United States	Houston	Texas	77041	Central
United States	Houston	Texas	77041	Central
United States	Naperville	Illinois	60540	Central
United States	Los Angeles	California	90049	West
United States	Melbourne	Florida	32935	South
United States	Eagan	Minnesota	55122	Central
United States	Eagan	Minnesota	55122	Central

1.4. Transactional and Measures Information Group

This is the “heart” of the dataset, containing the quantitative values generated from each sales order or business transaction:

- Order and date information: **Order_ID** and **Order_Date**, representing the unique order identifier and the time at which revenue was generated.
- Financial metrics: **Sales** (Revenue), **Profit**, **Quantity**, and **Discount**. These are quantitative fields that will be aggregated and calculated (such as Sum and Average) within the Fact Table for analytical reporting purposes.

Order_ID	Order_Date	Sales	Quantity	Discount	Profit
CA-2016-121755	1/16/2016	90.5700	3	0.0000	11.7741
US-2015-150630	9/17/2015	3083.4300	7	0.5000	-1665.0522
US-2015-150630	9/17/2015	9.6180	2	0.7000	-7.0532
US-2015-150630	9/17/2015	124.2000	3	0.2000	15.5250
US-2015-150630	9/17/2015	3.2640	2	0.2000	1.1016
US-2015-150630	9/17/2015	86.3040	6	0.2000	9.7092
US-2015-150630	9/17/2015	15.7600	2	0.2000	3.5460
CA-2017-107727	10/19/2017	29.4720	3	0.2000	9.9468
CA-2016-117590	12/8/2016	1097.5440	7	0.2000	123.4737
CA-2016-117590	12/8/2016	190.9200	5	0.6000	-147.9630
CA-2015-117415	12/27/2015	113.3280	9	0.2000	35.4150
CA-2015-117415	12/27/2015	532.3992	3	0.3200	-46.9764
CA-2015-117415	12/27/2015	212.0580	3	0.3000	-15.1470
CA-2015-117415	12/27/2015	371.1680	4	0.2000	41.7564
CA-2017-120999	9/10/2017	147.1680	4	0.2000	16.5564
CA-2016-101343	7/17/2016	77.8800	2	0.0000	3.8940
CA-2017-139619	9/19/2017	95.6160	2	0.2000	9.5616
CA-2016-118255	3/11/2016	45.9800	2	0.0000	19.7714
CA-2016-118255	3/11/2016	17.4600	2	0.0000	8.2062

2. Thiết kế mô hình vật lý (Physical Data Model Design)

Based on the results of the source data analysis, the physical data model is implemented using the Star Schema structure in order to optimize query performance and support efficient analytical processing. Instead of storing all information in a single flat table, the data is decomposed and reorganized into specialized tables that separate transactional data from descriptive data. This approach improves scalability, reduces redundancy, and enhances the performance of reporting and Business Intelligence operations.

In this design, the original flat dataset is transformed into five specialized tables, including one central Fact Table and multiple surrounding Dimension Tables. The Fact Table stores quantitative transactional data such as revenue, profit, quantity, and discount values, while the Dimension Tables contain descriptive information related to customers, products, geographic locations, and time. By organizing the data in this manner, the system can perform multidimensional analysis more efficiently and support complex reporting requirements within Power BI and other analytical platforms.

2.1. Customer Dimension Table (Dim_Customer)

Role: The Dim_Customer table is responsible for managing all customer identification information and segmentation attributes within the Data Warehouse. It stores descriptive customer data such as customer names, customer IDs, and customer segments, which are essential for customer-oriented analysis.

Benefits: This table enables businesses to track Customer Lifetime Value (CLV), analyze customer purchasing behavior, and evaluate the effectiveness of marketing strategies for different customer segments. By centralizing customer information into a dedicated dimension table, the system also improves data consistency and simplifies customer-related reporting.

2.2. Product Dimension Table (Dim_Product)

Role: The Dim_Product table stores all product-related information and product categorization data. It contains attributes such as product name, category, and sub-category, which provide context for transactional sales data.

Benefits: This structure standardizes product naming conventions and product classifications across the entire system. When changes occur, such as updating a category name or correcting product information, modifications only need to be applied in a single location instead of across millions of transaction records. This significantly improves maintainability and reduces data inconsistency.

2.3. Location Dimension Table (Dim_Location)

Role: The Dim_Location table contains geographic and regional attributes such as country, region, state, city, and postal code. It provides the spatial context necessary for geographic analysis and location-based reporting.

Benefits: Separating geographic information into a dedicated dimension table optimizes data filtering and analytical operations based on location. In addition, this design reduces the storage size of the Fact Table because the Fact Table only needs to store a compact Location_Key instead of repeatedly storing long textual address information for every transaction. This contributes to better storage efficiency and faster query performance.

2.4. Time Dimension Table (Dim_Time)

The Dim_Time table is a specialized component in the Data Warehouse that does not exist directly in the original CSV dataset but is generated from the Order_Date field during the ETL process.

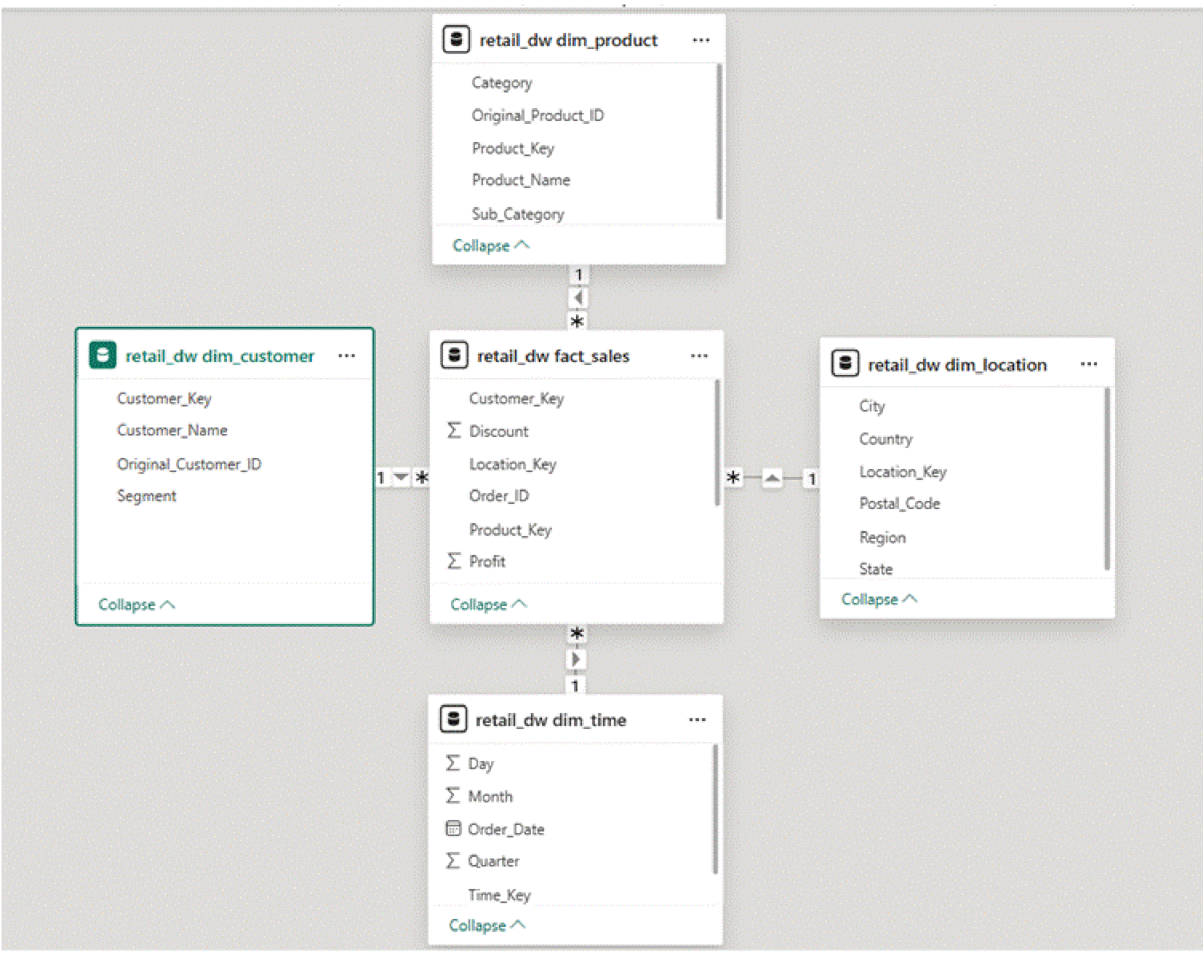
Hierarchical Levels: The table contains multiple time hierarchies, including Day, Month, Quarter, and Year, allowing users to analyze business performance at different levels of time granularity.

Importance: The Time Dimension is essential for time-series analysis and trend comparison. It enables advanced analytical operations such as Year-over-Year (YoY) revenue comparison, quarterly performance evaluation, and seasonal sales analysis. These types of analyses are difficult to perform efficiently using ordinary date fields alone.

2.5. Sales Fact Table (Fact_Sales)

Role: The Fact_Sales table is the central table of the Star Schema and stores all foreign keys referencing the four Dimension Tables, along with raw financial metrics generated from business transactions.

Characteristics: This is the largest table in the Data Warehouse, typically containing millions of transaction records. To maximize query and calculation performance, the Fact Table stores only numerical measures and foreign key IDs rather than descriptive information such as customer names or product names. This optimized structure enables faster aggregations, more efficient analytical queries, and improved performance for dashboards and reporting systems in Power BI.


retail_dw dim_product
Category
Original_Product_ID
Product_Key
Product_Name
Sub_Category
Collapse
retail_dw dim_customer
Customer_Key
Customer_Name
Original_Customer_ID
Segment
Collapse
retail_dw fact_sales
Customer_Key
Discount
Location_Key
Order_ID
Product_Key
Profit
Collapse
retail_dw dim_location
City
Country
Location_Key
Postal_Code
Region
State
Collapse
retail_dw dim_time
Day
Month
Order_Date
Quarter
Time_Key
Collapse

III. BUILDING THE ETL PIPELINE ON MYSQL

The ETL (Extract – Transform – Load) process is the backbone of every Data Warehouse system. In this project, the entire ETL workflow is implemented using SQL on the MySQL database management system, ensuring high consistency, reliability, and processing performance. The ETL process plays a critical role in transforming raw operational data into structured and analysis-ready information suitable for Business Intelligence applications.

By implementing the ETL pipeline directly in MySQL, the system can efficiently manage large volumes of transactional data while maintaining data integrity throughout the extraction, transformation, and loading stages. SQL-based processing also provides better control over data validation, cleansing, and standardization procedures, which are essential for building a reliable Data Warehouse. Furthermore, using SQL for ETL operations improves scalability and allows complex data transformations to be executed efficiently within the database environment, reducing processing overhead and enhancing overall system performance.

1. Staging Phase (Raw Data Loading and Staging Area)

The first stage of the ETL pipeline is to extract data from the **superstore.csv** file and load it into an intermediate table called **staging_table**. This stage serves as the initial entry point for raw data before it undergoes cleansing, transformation, and integration into the Data Warehouse.

To perform this process, the project utilizes the **Table Data Import Wizard** available in MySQL Workbench. This method provides an efficient and user-friendly solution for mapping columns from the CSV file into the initial database table structure without requiring complex manual data-loading commands. By using this tool, the import process becomes faster, more reliable, and less error-prone, especially when handling large datasets.

The staging table plays an important role as a temporary buffer zone between the original data source and the final Data Warehouse. It acts as a protective layer that preserves the integrity of the original dataset while allowing extensive data transformation and cleaning operations to be performed safely within the database environment. This approach ensures that any transformation errors or experimental modifications do not directly affect the source file.

During the staging phase, all imported fields, including numerical and date-related data, are typically stored in text format such as VARCHAR. This strategy helps prevent data loading failures caused by incompatible formats or inconsistent source data. Data type standardization and validation are then performed in later transformation stages where the data can be processed more carefully and systematically.

In addition, storing the raw data inside a database table significantly improves processing efficiency. Querying data directly from a database table is much faster and more efficient than repeatedly reading data from external CSV files. This optimization enhances ETL performance, reduces processing time, and creates a more stable environment for large-scale data transformation operations within the Data Warehouse system.

2. Advanced Transform and Load Process

After the data has been loaded into the Staging Area, the system begins executing complex transformation logic to convert the data from a flat-table structure into a standardized Star Schema model. This stage is one of the most important parts of the ETL pipeline because it ensures that raw operational data is cleaned, organized, and optimized for analytical processing within the Data Warehouse.

During the transformation process, the system applies various SQL operations to restructure the dataset into separate Fact Tables and Dimension Tables. Data cleansing procedures are also performed to remove duplicate records, handle missing or inconsistent values, standardize formats, and ensure overall data quality. In addition, business rules and derived calculations are applied to generate meaningful analytical attributes and metrics required for reporting and Business Intelligence analysis.

Once the transformation stage is completed, the processed data is loaded into the final Data Warehouse tables. Dimension Tables are populated first to establish descriptive business contexts, followed by the loading of transactional data into the Fact Table using the corresponding foreign keys. This structured loading process ensures referential integrity, improves query performance, and prepares the data for efficient multidimensional analysis in Power BI and other analytical platforms.

2.1. Advanced Date Transformation and Standardization

One of the biggest challenges in the retail dataset is that the **Order_Date** and **Ship_Date** columns are stored as strings in the **MM/DD/YYYY** format. In this format, MySQL cannot directly perform time-based operations such as calculating monthly revenue or comparing yearly sales performance.

- Applied technique: Using the **STR_TO_DATE()** function.
- Execution logic: The system parses the string values, identifies the month (**%m**), day (**%d**), and year (**%Y**) components, and converts them into the standard ISO date format (**YYYY-MM-DD**).
- The standardized date values form the foundation for extracting smaller temporal components such as Day, Month, Quarter, and Year, which are then used to populate the **Dim_Time** table in the Data Warehouse.

2.2. Deduplication and Dimension Standardization Logic

A key issue with flat data structures is that customer or product information is repeated thousands of times. For example, if a customer places 10 orders, the same customer name and ID will appear in 10 separate rows, even though it represents a single entity.

- Solution: Using a combination of the **INSERT INTO ... SELECT DISTINCT** statement.
- Execution process:
+ The system extracts a distinct set of attributes (e.g., Customer_Name, Segment) from the Staging table to ensure that only unique records are selected for each dimension. This step eliminates redundancy and prepares clean reference data for the Dimension Tables.
+ The deduplicated dataset is then inserted into the corresponding Dimension Table, ensuring that each business entity (such as a customer or product) is represented only once in the Data Warehouse.
+ MySQL's **AUTO_INCREMENT** feature is leveraged to automatically generate **Surrogate Keys** for each record in the Dimension Tables. These surrogate keys are simple integer values that replace natural business keys, significantly improving storage efficiency and optimizing JOIN performance between Fact and Dimension tables compared to using textual or business-based identifiers.

2.3. Loading Data into the Fact Table (Data Integration & Loading)

This is the final and most complex step of the ETL process. The **Fact_Sales** table does not directly retrieve data from the Staging table in a simple way; instead, it requires joining the staging data with all previously loaded Dimension Tables.

- Multi-layer Join Logic:
  + An **INNER JOIN** is performed between the Staging table and **Dim_Product** using the product name as the matching key, in order to retrieve the corresponding **Product_Key**.
  + The process continues by joining the result with **Dim_Customer** to obtain the **Customer_Key**, ensuring that each transaction is correctly mapped to its associated customer dimension.
  + The same joining strategy is then applied sequentially for **Dim_Location** and **Dim_Time**, allowing the system to fully replace all descriptive attributes with their corresponding surrogate keys.
- **Result:** The resulting Fact Table becomes a compact dataset consisting of numerical measures (Sales, Profit, Quantity) combined with surrogate identifiers (Keys). This structure is highly optimized and lightweight, enabling Power BI to perform aggregation and analytical calculations across millions of rows of data in a matter of seconds.

IV. DATA VISUALIZATION ON POWER BI

After the data has been standardized and loaded into the MySQL Data Warehouse, the final and most important step is to transform these raw numbers into a visual and interactive interface in the form of a Dashboard. This stage focuses on enabling business managers to understand the overall health and performance of the organization within just a few seconds.

1. Data Connectivity and Data Modeling Process

The connection between Power BI and MySQL is the initial step that requires a high level of precision in data structure to ensure the integrity and reliability of the final reports.

The connection method is established using **MySQL Connector/NET**, which enables a direct data link between Power BI and the MySQL database. In this project, the **Import mode** is selected instead of DirectQuery. This approach allows data to be loaded into Power BI's in-memory engine, making report interactions significantly faster and more responsive when users apply filters or interact with visualizations.

After establishing the connection, the next step is to construct the **Star Schema model** within Power BI's Model View. In this stage, all tables are organized logically into a star-like structure, with the Fact table positioned at the center and Dimension tables surrounding it.

However, this process can introduce certain challenges. Power BI may automatically detect relationships based on column names, which can lead to issues such as **circular dependencies** or **ambiguous relationships** between tables. These issues can negatively impact data accuracy and report behavior if not properly controlled.

To resolve this, relationships are manually defined using key columns such as **Product_Key**, **Customer_Key**, **Location_Key**, and **Time_Key**, ensuring that each Dimension table is correctly linked to the Fact table. All relationships are strictly enforced as **one-to-many (1:N)** with a single-direction filter flow from Dimension tables to the Fact table. This design guarantees a clean, stable, and predictable data model, which is essential for accurate analytics and high-performance dashboard execution.

2. Building Advanced Analytical Metrics (DAX Measures)

The DAX (Data Analysis Expressions) language is considered the “soul” of Power BI, as it enables dynamic calculations that are difficult to express flexibly using pure SQL, especially within an interactive reporting interface.

2.1. Nhóm chỉ số Doanh thu và Lợi nhuận

These are the most fundamental yet essential indicators used to evaluate the overall scale of the business:

- Total Sales: This measure calculates the total revenue generated from all transaction records in the invoice dataset.
  + *Formula:* Total Sales = SUM('retail_dw fact_sales'[Sales])
- Total Profit: This measure represents the total profit after accounting for costs and discounts across all transactions.
  + *Formula:* Total Profit = SUM('retail_dw fact_sales'[Profit])

2.2. Efficiency Measures Group

These ratio-based metrics are used to evaluate business performance quality rather than relying solely on absolute values. They provide deeper insight into operational efficiency and profitability.

- **Profit Margin (%)** represents the ratio of profit to total revenue. It indicates how much profit is generated from each unit of revenue, serving as a key indicator of overall business efficiency.
  + *Formula:* Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)
- **Average Order Value (AOV)** measures the average value of each transaction. This metric helps identify customer spending behavior and supports the segmentation of target customer groups for more effective marketing and sales strategies.

2.3. Growth Indicators Group (Time Intelligence)

Comparing performance across different time periods is one of the most powerful capabilities of a BI system. These time intelligence metrics allow the business to track trends, identify seasonal patterns, and evaluate performance changes over time in a structured and meaningful way.

Sales Last Month (SML) represents the total revenue generated in the previous month. This measure provides a baseline for short-term performance comparison and helps managers understand recent business outcomes in context.

Month-over-Month (MoM) Growth measures the percentage change in revenue compared to the previous month. It highlights whether the business is growing or declining on a monthly basis, making it a critical early-warning indicator for performance shifts.

Business Impact: These metrics enable managers to quickly detect revenue drops or growth slowdowns and take timely corrective actions. By continuously monitoring time-based performance trends, businesses can make more informed strategic decisions and improve overall operational responsiveness.

3. Dashboard Design and User Experience (UX/UI)

The dashboard is designed based on the principle of "from overview to detail," ensuring that users are not overwhelmed by too much information at once.

3.1. Executive Summary Layer

Located at the top of the dashboard, this section is designed as the **Executive Summary Layer**, typically implemented using Multi-row Cards or KPI Cards.

Purpose: It presents the most critical business indicators in a concise and highly visible format, such as Total Sales, Total Profit, Total Orders, and Growth Rate. These metrics are carefully selected to represent the overall health and performance of the business.

Impact: This layer allows executives and managers to quickly grasp the current business situation within the first few seconds of viewing the dashboard. By summarizing key performance indicators in a clear and visually prominent way, it significantly reduces the time needed for decision-making and improves situational awareness at the leadership level.

3.2. Temporal Analysis Layer

Line charts or area charts are used in this section.

Content: It visualizes revenue fluctuations by month, quarter, and year.

Interactive features: The dashboard integrates a drill-down capability, allowing users to click on a specific year to view monthly details, and further click on a month to explore daily performance. This enables the identification of market seasonality patterns in the retail business.

3.3. Categorical & Structural Analysis Layer

Stacked bar charts and donut charts are used in this section to analyze the structural composition of the business data.

Product-based analysis: This view compares revenue across major product categories such as Technology, Furniture, and Office Supplies, helping identify which segments contribute most to overall sales performance.

Geographical analysis: Map visuals or bar charts are used to rank states and cities by revenue, providing clear insights into regional performance differences and highlighting high-performing locations.

Dynamic filters (Slicers): The dashboard includes interactive slicers that allow users to filter all visuals by Region, Customer Segment, or Fiscal Year. When a user selects a specific region or segment, all related charts automatically update in real time, transforming the dashboard into a fully interactive and "live" analytical report.

V. Business Performance Analysis

After the dashboard system has been completed, interpreting the data becomes the most important step in transforming numbers into actionable business decisions. Based on the established Star Schema model, multidimensional analysis is performed to uncover hidden patterns within the Superstore retail market.

1. Time-Series Trend Analysis

Using the **Dim_Time** table linked to the **Fact_Sales** table, the line chart clearly illustrates the business performance pattern over time.

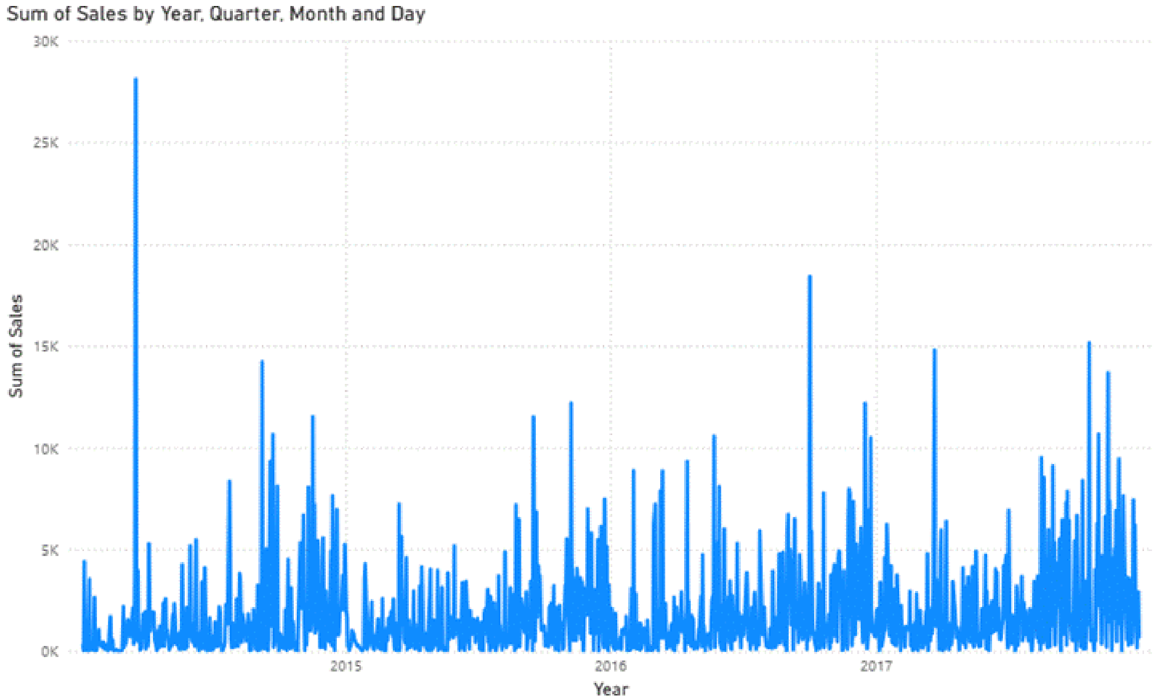
Pattern identification: Revenue is not evenly distributed across months but shows a significant surge toward the end of the year, particularly in November and December. During this period, Month-over-Month (MoM) growth can reach 1.5 to 2 times higher compared to the early months of the year.

External factors: This spike is mainly driven by major global shopping events such as Black Friday, Christmas, and New Year. During these periods, consumer demand increases sharply across all product categories, as customers purchase goods for gifting and home decoration purposes.

Internal factors: The company also actively intensifies marketing campaigns and applies strong discount strategies during this period to stimulate demand and clear out remaining inventory before year-end.

Strategic recommendations:

- **Supply chain management:** Inventory should be prepared in advance from the end of Q3 to avoid stockouts during the peak season in December.
- **Workforce optimization:** Additional staffing should be allocated to logistics and customer service teams during peak months to handle increased order volume and maintain service quality.



2. Product Performance Analysis

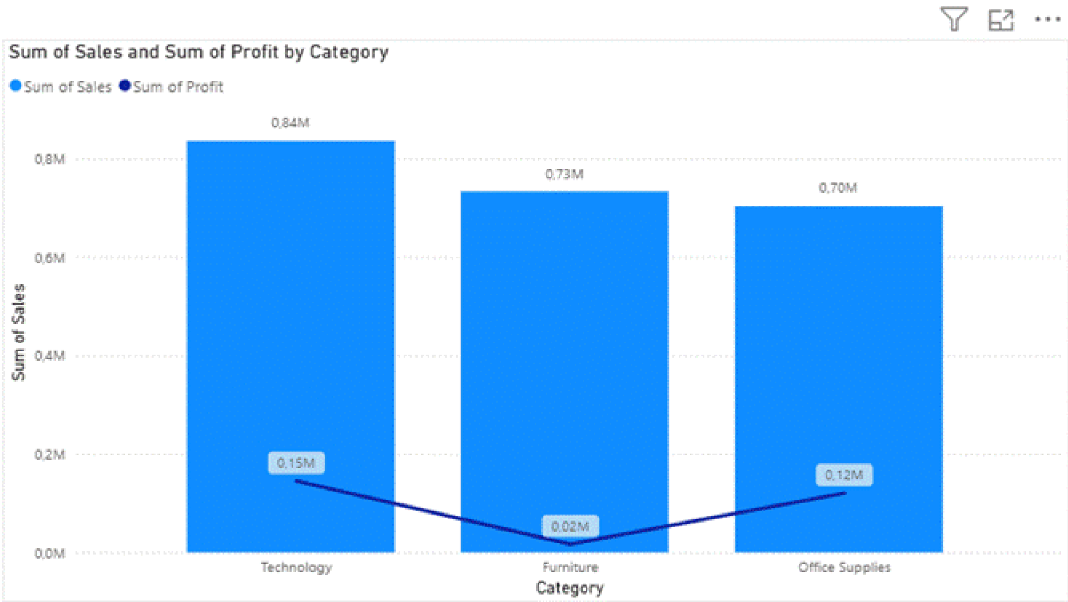
The integration between **Dim_Product** and the Fact table reveals a clear divergence between total revenue and order volume across different product groups.

Technology category: This segment generates the highest revenue in the system. Although it does not always lead in terms of quantity sold, its high unit price (e.g., smartphones, computers, and other electronic devices) ensures that total sales value remains dominant. However, this category also carries certain risks, as profit margins tend to fluctuate due to intense market competition and high after-sales costs such as warranty and technical support.

Office Supplies category: While this segment does not contribute the highest revenue, it demonstrates the most stable order volume and purchase frequency. It plays a crucial role as a consistent cash-flow generator, supporting daily operations and encouraging repeat customer purchases, which strengthens long-term customer retention.

Furniture category: This group shows relatively strong revenue performance but tends to generate lower profit margins. The main reasons include high logistics costs due to bulky products and a higher return rate compared to other categories.

Strategic implication: The Office Supplies segment can be strategically used as a "loss leader" or entry product group to support cross-selling initiatives, encouraging customers to purchase higher-value Technology products and thereby increasing overall profitability.



3. Geographic Performance Analysis

Data from the **Dim_Location** table enables managers to clearly observe performance disparities across different regions.

Central Region:

This region contributes a significant proportion of total revenue, accounting for approximately 40% of the overall system sales. It represents a key market with a large and dense customer base.

However, a notable paradox emerges in terms of profitability: despite generating the highest revenue, this region records the lowest profit margin compared to other areas.

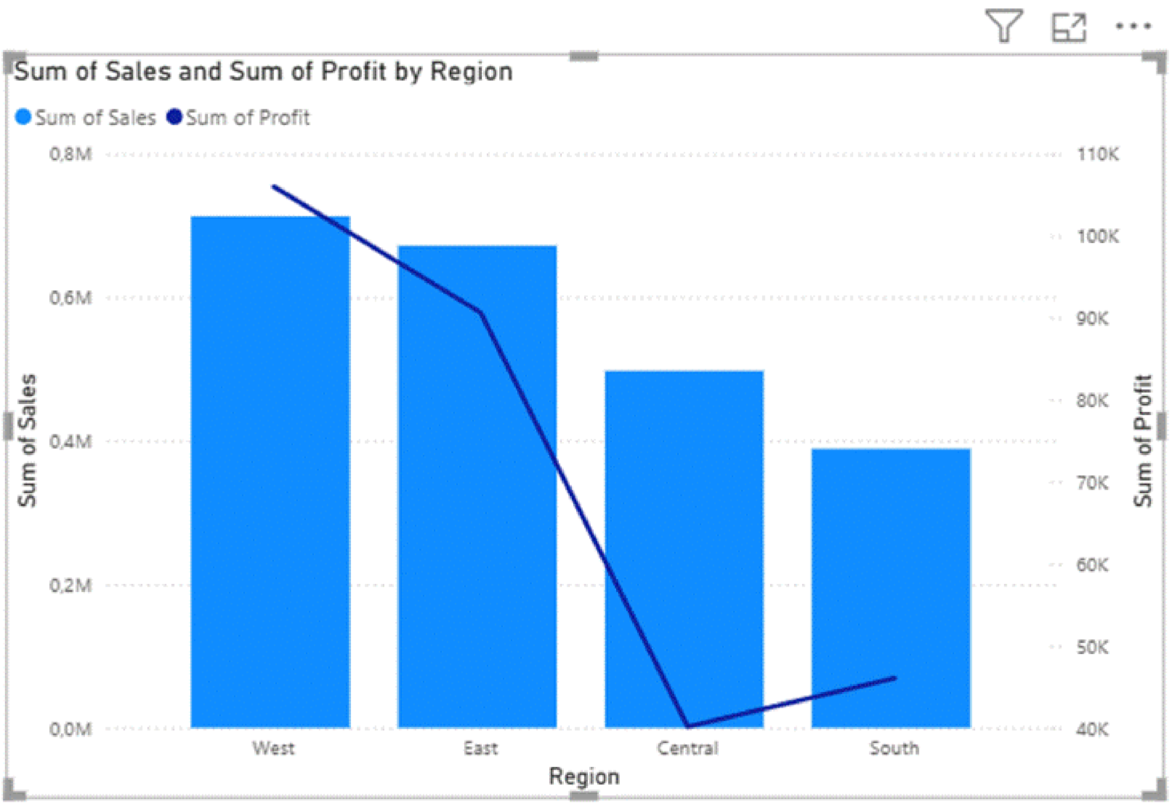
Root cause analysis:

- **Intense competition:** The Central Region is highly competitive, forcing the company to frequently apply heavy discounts to maintain market share.

- **Operational costs:** Warehousing and logistics costs in major urban areas are significantly higher, directly reducing net profit margins.

Strategic recommendations:

- **Pricing optimization:** Discount policies, especially those exceeding 20%, should be reviewed carefully, as analysis indicates that some transactions in this region even result in negative profit due to excessive price reductions.
- **Market rebalancing:** Marketing resources should be partially shifted toward regions with lower revenue but higher profit margins (such as the West or South) in order to achieve a more balanced and sustainable overall profit structure.

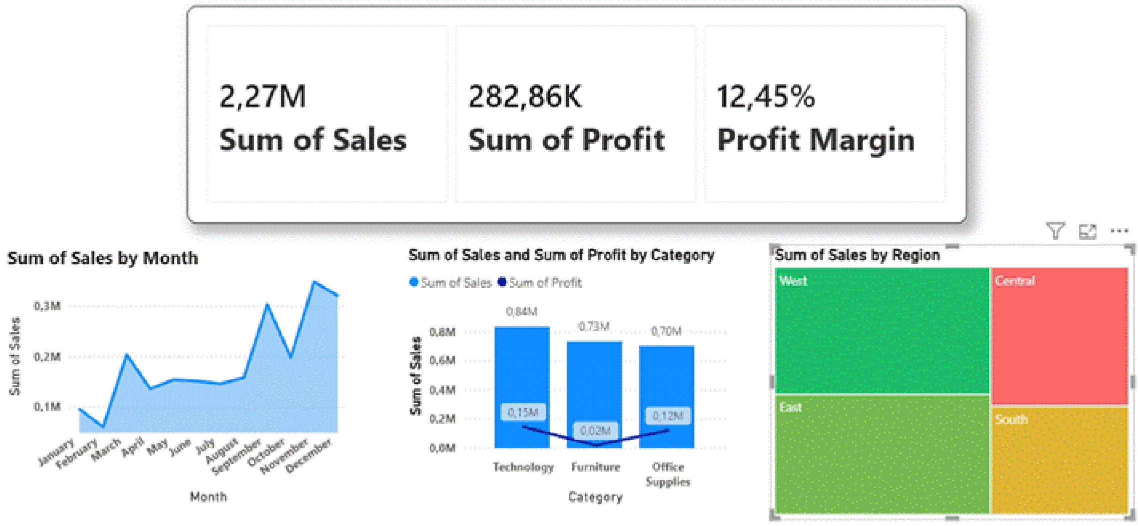


4. Summary Evaluation

The multidimensional analysis conducted through the Data Warehouse demonstrates that high revenue does not necessarily translate into high profit. The BI system has successfully uncovered critical "blind spots" that were not clearly visible in traditional flat Excel-based reporting.

Through structured analysis across products, regions, and time dimensions, it becomes evident that sustainable business performance depends not only on sales volume but also on margin optimization and operational efficiency. In particular, improving the balance between product category performance and refining discount strategies by geographic region are key factors for long-term profitability.

Overall, the BI system provides a more comprehensive and accurate understanding of business performance, enabling Superstore to move toward more data-driven decision-making and sustainable growth in the coming years.



VI. CONCLUSION AND RECOMMENDATIONS

1. Project Conclusion

The "Retail Sales Business Intelligence System" project has successfully completed the full end-to-end data pipeline, from raw data extraction to information visualization, establishing a solid foundation for digital business management.

1.1. Technical and Architectural Achievements

Structure optimization: The system successfully transformed a flat-table structure into a **Star Schema model**. Centralizing data within the **Fact_Sales** table eliminates redundancy and significantly improves query performance for analytical reporting.

Standardized ETL process: A structured ETL pipeline was implemented on MySQL, effectively handling data format inconsistencies and duplicate records. The use of **Surrogate Keys** ensures data integrity and supports future system scalability.

1.2 Business Value and Decision Support

Multidimensional visualization: Power BI provides a comprehensive view from high-level KPIs to detailed breakdowns by product, customer, and region, enabling real-time business performance monitoring.

Risk detection: The system uncovers hidden issues such as regions with high revenue but negative profit caused by excessive discounting, which were previously difficult to detect in flat reporting systems.

Automation: Manual reporting processes have been replaced with automated, near real-time dashboards, allowing faster and more accurate responses to market changes.

2. Recommendations and Future Development

To progress toward a fully **data-driven business model**, the system should be enhanced in the following directions:

Inventory management integration: Incorporating inventory data will enable analysis of stock turnover, optimization of storage costs, and prevention of stockouts for key products.

Machine Learning applications: Implement forecasting models to predict revenue for the next 6–12 months, supporting better budget planning and supply chain decisions.

Customer segmentation (RFM model): Deploy RFM analysis within Power BI to identify loyal customers and at-risk customers, enabling more personalized marketing strategies.

Cloud migration: Move the system to cloud platforms such as **Google BigQuery** or **Azure SQL** to enhance security, scalability, and remote accessibility across departments.